

Lodz University
of Technology

Final project

Equipment Design and Material Engineering

Design and Realization of a Clap Switch PCB

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Contents

1 Preliminary CAD Design Activity: Multivibrator **3**

1.1 EasyEDA 3

1.2 KiCad 5

1.3 Altium 5

1.4 Comparison of the Design Tools 6

2 Introduction **7**

3 Goals **8**

4 Functional Analysis of the Circuit **9**

5 Realization of the Schematic **10**

6 Selecting components, packages and footprints **12**

7 Cost Estimate **13**

8 PCB Design **14**

9 Assembly and Testing **16**

10 Conclusions **19**

1 Preliminary CAD Design Activity: Multivibrator

Before starting the development of the final Clap Switch project, a preliminary laboratory activity was carried out in order to become familiar with the complete PCB design workflow and with the three CAD tools introduced during the course: EasyEDA, KiCad and Altium Designer.

The exercise consisted of designing a multivibrator circuit and implementing it as a printed circuit board. The aim of this activity was not only to reproduce a simple electronic circuit, but also to understand the main phases of electronic design: project creation, schematic capture, component selection, footprint assignment, transfer to the PCB editor, component placement, routing and final layout verification.

This preliminary activity was performed using EasyEDA, KiCad and Altium Designer. The same multivibrator circuit was recreated in the three software environments in order to compare their design workflows and understand the differences between them.

1.1 EasyEDA

EasyEDA is the CAD tool that was also used to implement the final Clap Switch project. The workflow started with the creation of a new project and the drawing of the multivibrator schematic, which is the same for both EasyEDA projects, what changes is the types and footprints of the components used.

The circuit is composed of two NPN transistors, four resistors and two capacitors arranged in a cross-coupled configuration. This circuit continuously alternates between two states, generating an oscillating signal without requiring an external clock source. The schematic used for the exercise is shown in Figure 1.

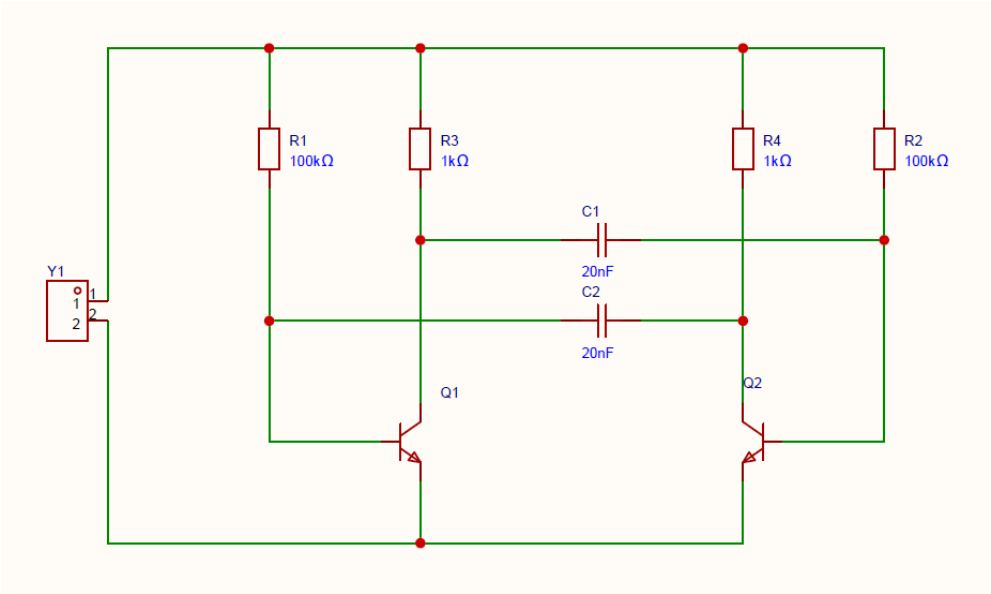
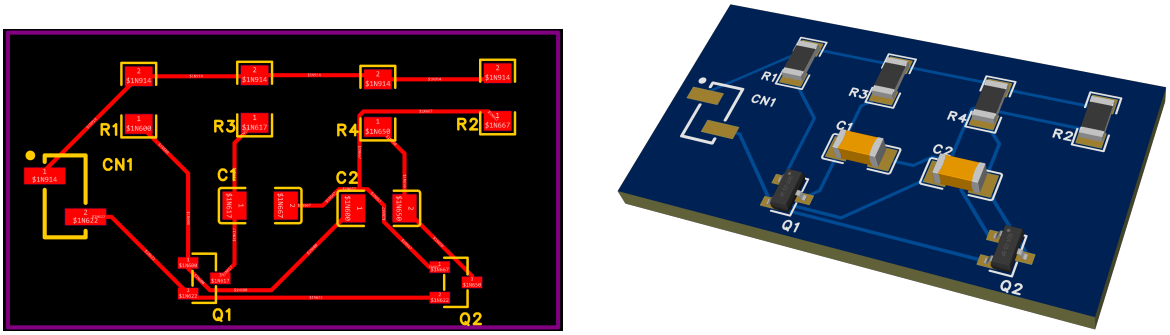


Figure 1: Multivibrator schematic created in EasyEDA.

After completing the schematic, suitable footprints were assigned to the components. Two different PCB implementations were then developed for operation: one using Surface Mount Technology (SMT) which was routed mainly on the top copper layer and one using Through-Hole Technology (THT) which was routed mainly on the bottom layer.

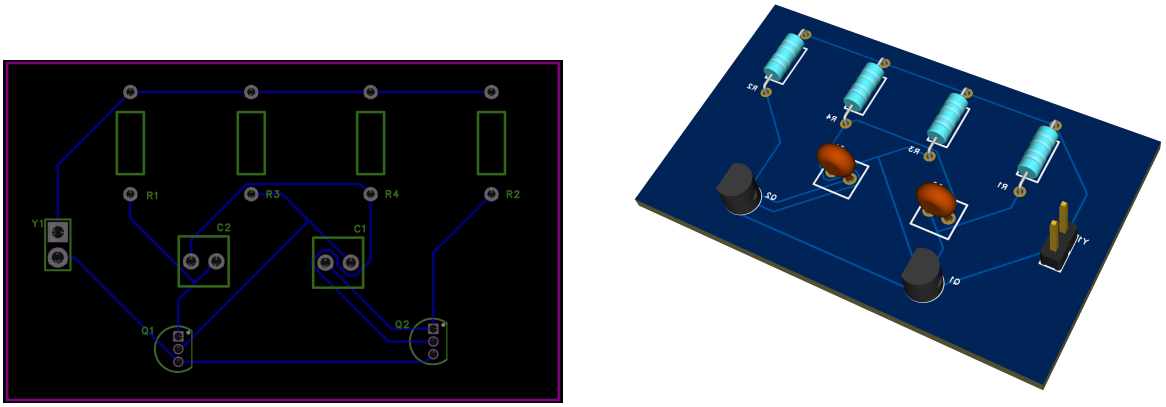
In the SMT version, surface-mount components were used, including SOT-23 transistors and SMD resistors and capacitors. This solution allowed a compact PCB layout, and the routing was mainly performed on the top copper layer. The corresponding PCB layout and 3D view are shown in Figure 2.



(a) SMT PCB layout. *(b) SMT 3D view.*

Figure 2: SMT implementation of the multivibrator in EasyEDA.

In the THT version, through-hole components were used. This implementation required a larger board area because of the physical dimensions of the components and the need for drilled holes. In this case, the routing was mainly performed on the bottom copper layer. The PCB layout and the corresponding 3D view are shown in Figure 3.



(a) THT PCB layout. *(b) THT 3D view.*

Figure 3: THT implementation of the multivibrator in EasyEDA.

The EasyEDA activity was particularly useful because the schematic editor, footprint assignment and PCB editor are strongly integrated. This made the transition from the circuit diagram to the PCB layout clear and intuitive. Moreover, the 3D visualization tool allowed an immediate check of the final board appearance and of the component placement.

1.2 KiCad

The same multivibrator project was also recreated in KiCad during the laboratory sessions. The design workflow followed the standard sequence used in PCB development: project creation, schematic capture, footprint assignment, transfer to the PCB editor, component placement and routing.

Compared with EasyEDA, KiCad required a more structured approach. The schematic editor and PCB editor are managed as separate environments, and for this reason particular attention was needed when associating schematic symbols with the corresponding physical footprints. This made the process slightly less immediate at the beginning, but it was useful for understanding more deeply the relationship between the electrical schematic and the physical PCB implementation.

The activity with KiCad provided experience with an open-source EDA environment and helped to understand how libraries, footprints and PCB rules are managed in a more manual and flexible design workflow.

1.3 Altium

The multivibrator circuit was also implemented using Altium Designer. This represented the first contact with a professional PCB design environment.

The workflow included project creation, schematic drawing, component annotation, footprint association, transfer to the PCB document and routing. Altium Designer appeared more complex than EasyEDA and KiCad at the beginning, but it also provided a more complete and integrated environment.

In particular, Altium Designer showed advanced tools for project organization, design-rule checking and PCB layout management. This made it possible to understand the type of workflow commonly used in industrial electronic design.

1.4 Comparison of the Design Tools

The preliminary activity made it possible to compare the three CAD environments by applying them to the same circuit. Although the final objective was identical in all cases, the user experience and the organization of the design flow were different.

Table 1: Comparison of the CAD tools used during the preliminary multivibrator activity.

Tool	**Main characteristics**	**Personal experience during the activity**
EasyEDA	Web-based environment with integrated schematic editor, footprint assignment, PCB editor and 3D visualization. It allows a fast transition from schematic to PCB layout.	It was the most intuitive tool and the easiest to use during the first design activities. For this reason, it was later selected for the development of the final Clap Switch PCB.
KiCad	Open-source desktop software with separate schematic and PCB environments. It offers good flexibility and detailed control over libraries, symbols and footprints.	It required a more structured workflow than EasyEDA, especially during symbol-to-footprint association. This made it useful for understanding the PCB design process in greater detail.
Altium Designer	Professional PCB design software with advanced project management, design-rule checking and layout tools. It is closer to industrial electronic design workflows.	It was the most complete environment, but also the most complex at the beginning. It helped to understand how professional PCB projects are organized and verified.

From this comparison, EasyEDA resulted as the most immediate and practical software for a first PCB design experience. KiCad provided greater flexibility and a better understanding of the internal structure of a PCB project, while Altium Designer offered the most advanced and professional environment.

Based on the experience gained during the course, EasyEDA was selected for the development of the final Clap Switch PCB. Its simplicity, integrated component libraries, direct schematic-to-PCB workflow and 3D visualization tools made it the most suitable choice for completing the final project within the available time. The experience acquired with KiCad and Altium Designer was nevertheless important, because it allowed a broader understanding of modern PCB design methodologies and of the differences between educational, open-source and professional CAD environments.

2 Introduction

This project was developed within the course *Equipment Design and Material Engineering* with the objective of applying in a practical way the entire process of development of an electronic device, from the definition of the initial idea to the physical realization and final testing of the system.

The activity required the analysis of functional requirements, the selection of electronic components, the design of the circuit diagram, the creation of the printed circuit board (PCB) using EasyEDA software, the production of the board and the subsequent assembly of the electronic components. Finally, the circuit was subjected to a testing phase to verify its correct functioning and correspondence to the initially defined specifications.

Needed. Every project, as the professor said in class, arises from a need and we engineers have the task of devising something. The idea behind the project was born from a practical need encountered during the stay at the BaseCamp university dormitory. The room lighting control is in fact positioned at a distance that makes it difficult to turn the light on and off directly from the bed.

Idea. From this observation came the idea of developing a system that would allow remote lighting to be controlled through a simple sound control, eliminating the need to physically reach the switch.

To meet this need, it was chosen to create a *Clap Switch* type circuit, that is, a system capable of detecting clapping hands and using it as a command to activate or deactivate an electrical load.

3 Goals

The main objective of the project was to design, manufacture and test an electronic system capable of controlling the switching on and off of an electrical load by means of a sound command generated by clapping hands.

The specific objectives of the work were:

- analyze the problem to be solved and define the functional specifications of the system;
- design the wiring diagram using EasyEDA software;
- identify the real components available on the market and associate the correct footprint with each component on the wiring diagram;
- create the printed circuit board (PCB) layout and order on the JLCPCB website;
- assemble the circuit by soldering the electronic components;
- check the correct functioning of the system;

At the end of the project, the system was found to be fully functional, allowing the control of an electrical load through clapping hands and demonstrating the achievement of the set objectives.

4 Functional Analysis of the Circuit

Before going into the circuit details, I focused on creating a block diagram that represented the situation based on the circuit's objective.

Figure 4 shows the block diagram of the system, highlighting the main functions performed by the different stages of the circuit.

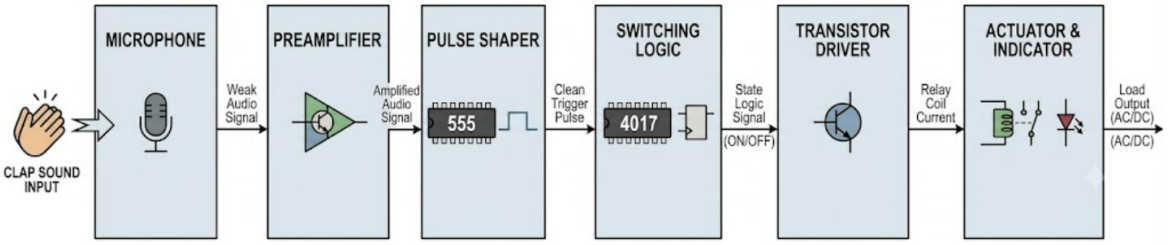


Figure 4: Clap Switch System Block Diagram.

The operation of the circuit can be divided into six main functional blocks.

The first block consists of the microphone, which has the task of converting the acoustic signal generated by clapping hands into a change in electrical voltage. Since the signal produced by the microphone has a very small amplitude, it is sent to a preamplification stage which increases its level making it suitable for subsequent processing.

The amplified signal is then processed by a pulse shaping stage (*pulse shaper*), whose task is to transform the acoustic signal, generally irregular and of variable duration, into a well-defined and reproducible electrical pulse. In this way the system is less sensitive to variations in the sound signal and more reliable during operation.

The generated pulse is then sent to the switching logic block (*switching logic*), responsible for alternating the ON and OFF states of the system. In fact, each clapping of hands causes the generation of a new pulse that determines the change in state of the circuit.

The output of the switching logic subsequently drives a transistor driver stage. This block is responsible for supplying the current necessary to activate the relay, while at the same time keeping the control logic isolated from the commanded load.

The last block includes the relay and the signal LED. The relay allows switching of the external load, while the LED provides a visual indication of the system status, allowing you to immediately check whether the circuit is in the active or inactive state.

The interaction between these blocks therefore allows a simple clap of hands to be transformed into a command to turn an electrical device on or off, meeting the functional objectives of the project.

5 Realization of the Schematic

Once the functional architecture of the system had been defined, the schematic was created using the EasyEDA software. During this phase, all the components necessary for the circuit's operation were inserted and the related electrical connections were made.

As an initial reference, a similar project available online *electronicshub* was analyzed, from which the general architecture of the circuit was taken. The components used were selected from those available in EasyEDA's libraries and from the suppliers consulted, favoring functionally equivalent models compared to those originally proposed. For example, some resistive values have been approximated to commercially available normalized values and the protection diode has been replaced with a component belonging to the same family of rectifier diodes. These modifications do not alter the operating principle of the circuit and fall within normal design tolerances.

Subsequently, the scheme was autonomously reconstructed within the EasyEDA environment, verifying the correct connection between the various functional blocks and its consistency with the operating principle defined in the analysis phase.

The final scheme includes an electret microphone for acoustic signal detection, two NPN transistors used for signal preamplification and relay driving respectively, a NE555 timer for control pulse generation, a CD4017 counter for ON/OFF switching management, a load control relay and a system status reporting LED.

Table 2 summarizes the main components used in the wiring diagram and the value or type.

Table 2: Components used in the Clap Switch circuit.

Component	**Value / Type**
Microphone (MIC1)	Electret microphone
Counter IC (IC1)	CD4017BE
Timer IC (IC2)	NE555P
NPN transistors (T1, T2)	BC547 (2 pcs)
Diode (D1)	1N4007
Relay (RELAY1)	12 V relay
LED (LED1)	Red LED
Resistors	100 Ω, 560 Ω, 4.7 kΩ, 33 kΩ, 18 kΩ (3 pcs)
Ceramic capacitors (C1, C2)	100 nF
Electrolytic capacitor (C4)	4.7 μF, 50 V
Connectors (J1, J2)	2-pin, 3-pin

At the end of the activity, the complete circuit diagram shown in Figure 5 was obtained, which formed the basis for the subsequent phases of package selection, footprint association and PCB design.

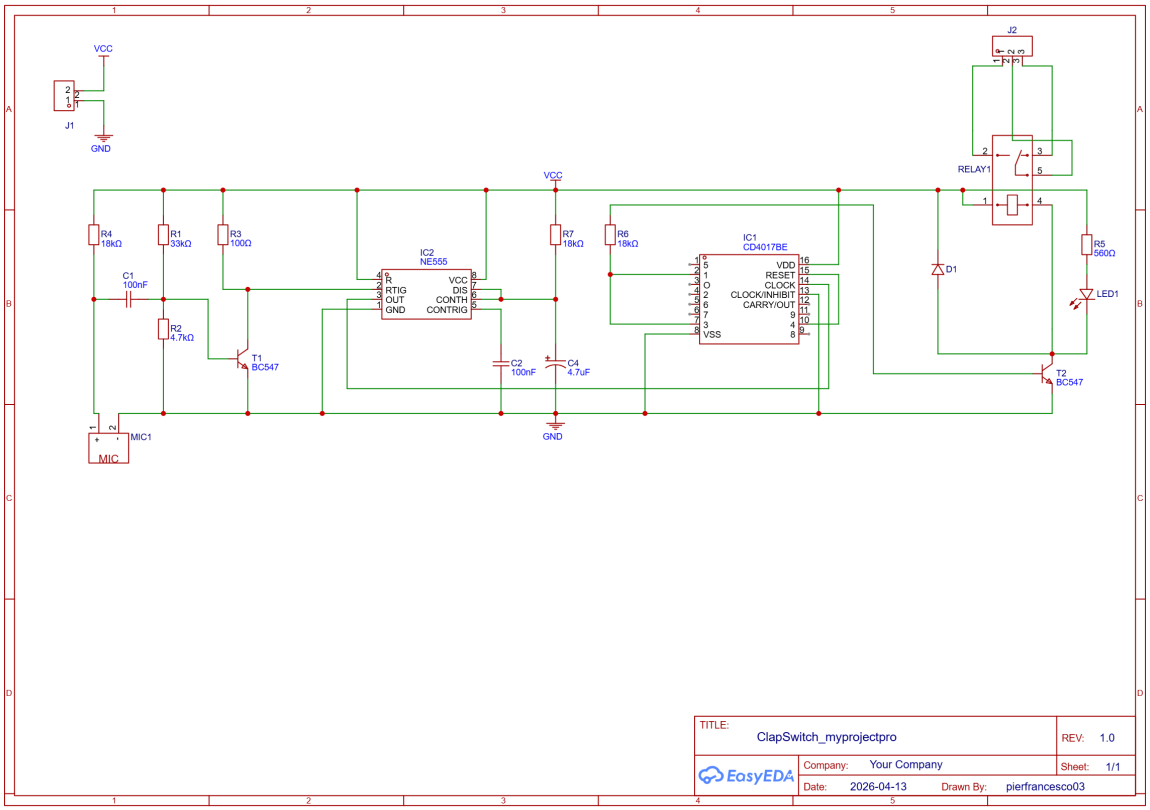


Figure 5: Diagram of the Clap Switch circuit made in EasyEDA.

6 Selecting components, packages and footprints

We now come to a fundamental step to which I have devoted a great deal of time. First of all I chose the type of components to use, I preferred Through-Hole Technology (THT) components. Once the circuit diagram was completed, it was necessary to select real components compatible with the physical construction of the printed circuit board. For this purpose, the TME catalogue was used as a reference to verify the availability of the components, the mechanical characteristics of the packages and the footprints.

For each component chosen on TME, the datasheet with the measurements was examined and then an appropriate footprint was associated on EasyEDA, verifying the correspondence between the package and the measurements declared in the datasheet on TME of the chosen component and the layout available within EasyEDA. This phase was particularly important because if you make the wrong choice of footprints for the components, you do not have correct positioning of the holes and pads on the PCB.

Table 3 reports the selected components and their footprints used during PCB design.

Table 3: Selected components and their footprints used in the project.

Reference	**Component**	**Footprint EasyEDA**
IC2	NE555P	PDIP-8_L9.5-W6.4-P2.54-LS7.6-BL
IC1	CD4017BE	DIP-16_16P-L20.2-W6.6-P2.54-LS7.6-BL
T1	BC547	TO-92-3_L4.9-W3.7-P1.27-L
T2	BC547	TO-92-3_L4.9-W3.7-P1.27-L
D1	1N4007 diode	DO-41_BD2.4-L4.7-P8.70-D0.9-RD
MIC1	Electret microphone	MIC-TH_BD6.0-P1.90-D0.7_EM-6050P
LED1	5 mm LED	LED-TH_BD5.0-P2.54-FD
RELAY1	12 V relay	RELAY-TH_SRD-XXVDC-XL-C
J1	2-pin connector	HDR-TH_2P-P2.54-V-M-3
J2	3-pin connector	HDR-TH_3P-P2.54-V-M-B
R1	33 kΩ resistor	RES-TH_BD2.5-L6.8-P10.00-D0.6
R2	4.7 kΩ resistor	RES-TH_BD2.5-L6.8-P10.00-D0.6
R3	100 Ω resistor	RES-TH_BD2.5-L6.8-P10.00-D0.6
R4	18 kΩ resistor	RES-TH_BD2.5-L6.8-P10.50-D0.5
R5	560 Ω resistor	RES-TH_BD2.5-L6.8-P10.50-D0.5
R6	18 kΩ resistor	RES-TH_BD2.5-L6.8-P10.50-D0.5
R7	18 kΩ resistor	RES-TH_BD2.5-L6.8-P10.50-D0.5
C1	100 nF ceramic capacitor	CAP-TH_L6.5-W4.5-P5.00-D0.6
C2	100 nF ceramic capacitor	CAP-TH_L6.5-W4.5-P5.00-D0.6
C4	4.7 μF electrolytic capacitor	CAP-TH_BD5.0-P2.00-D0.8-FD

7 Cost Estimate

After selecting the components and the corresponding footprints, a cost estimate was carried out in order to evaluate the economic feasibility of the Clap Switch prototype.

The Bill of Materials reports the components required for the realization of the circuit, while the cost estimate associates an approximate market price with each component or component group. The prices were estimated using the TME catalogue links collected during the component selection phase.

The estimate only considers the electronic components required for one assembled prototype. PCB manufacturing, shipping costs, taxes and laboratory tools are not included in this calculation.

Table 4: Estimated cost of the electronic components for one Clap Switch prototype.

Component	**Reference**	**Quantity**	**Estimated cost [USD]**
NE555P	IC2	1	0.49
CD4017BE	IC1	1	1.09
BC547	T1, T2	2	0.26
Microphone	MIC1	1	1.14
1N4007	D1	1	0.12
Relay 12 V	RELAY1	1	0.60
Red 5 mm LED	LED1	1	0.34
2-pin connector	J1	1	0.03
3-pin connector	J2	1	0.03
Resistors	R1–R7	7	0.24
100 nF ceramic capacitors	C1, C2	2	0.08
4.7 μF electrolytic capacitor	C4	1	0.03
Total estimated component cost			**4.45**

The estimated cost of the electronic components for one prototype is approximately 4.45 USD, which is well below the required limit of **100 PLN**. This confirms that the circuit is economically suitable for a small laboratory prototype. The most expensive components are the electret microphone, the CD4017BE counter and the relay, while passive components such as resistors and capacitors have a very small impact on the total cost.

8 PCB Design

After completing the wiring diagram and associating the footprints with the various components, the PCB was designed using EasyEDA's PCB Editor environment.

The initial phase consisted of automatically importing the wiring diagram into the PCB editor. Subsequently, the components were positioned, rotated and arranged on the board to obtain a simple and orderly layout, reducing the number of intersections and minimizing the length of the most critical connections. During this phase, reference was also made to the schematic to maintain an arrangement consistent with the functional structure of the circuit.

Once the arrangement of the components was defined, the routing of the tracks was carried out. At the end of the routing, it was verified that there were no unrouted connections (airwires) and that the layout was electrically complete. In particular, a width of 0.5 mm was chosen for most tracks, while the connections between the relay and the J2 connector, intended for interfacing with the external 230 V load, were made with a width of 1 mm to ensure greater mechanical and electrical robustness.

Four fixing holes were also inserted at the corners of the board, with a hole diameter of 3.2 mm and an external diameter of 3.5 mm, in order to allow the PCB to be mounted inside a container or on a mechanical support.

Figure 6 shows the final PCB layout made in EasyEDA, while Figure 7 shows the three-dimensional view of the board before production.

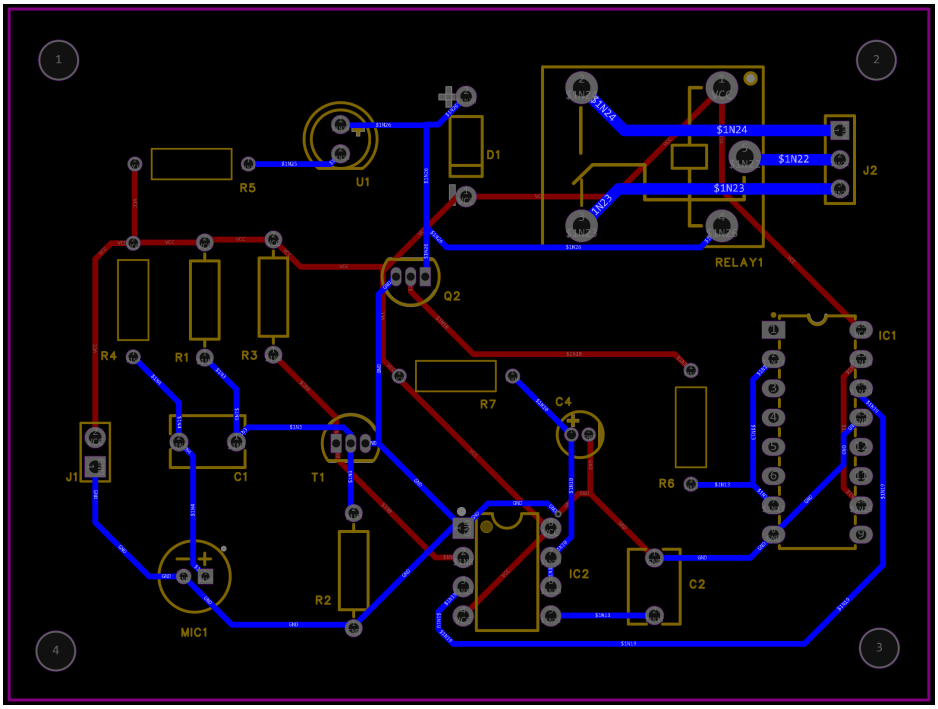


Figure 6: Final PCB layout made in EasyEDA.

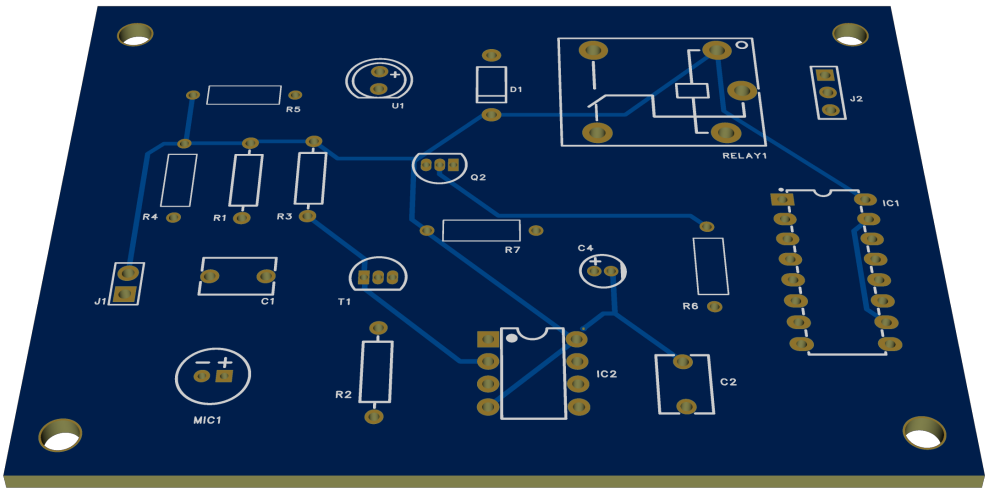


Figure 7: 3D view of the PCB.

Once the routing phase was completed and the layout errors were verified, the Gerber files needed to manufacture the board were generated. The PCB was later ordered through the JLCPCB website. Although the platform allows several manufacturing parameters to be customized, the standard options were selected because the circuit was intended to be used as a prototype and later placed inside a 3D-printed enclosure.

Five prototype boards were ordered for the project, sufficient for the planned assembly and testing activities.

9 Assembly and Testing

After receiving the produced PCB, the circuit was manually assembled during laboratory activities. Since all the components used are of the Through-Hole (THT) type, the assembly was carried out by inserting the terminals of the components into the respective holes provided on the printed circuit board.

Before starting the soldering operations, a careful component preparation phase was carried out. All the elements necessary for assembly were taken from the containers present in the laboratory and subsequently identified based on the nominal value and type. To avoid possible errors during assembly, the components were organized on a sheet of paper, reporting the relative reference and value for each, paying particular attention to the resistors, which could easily be confused with each other.

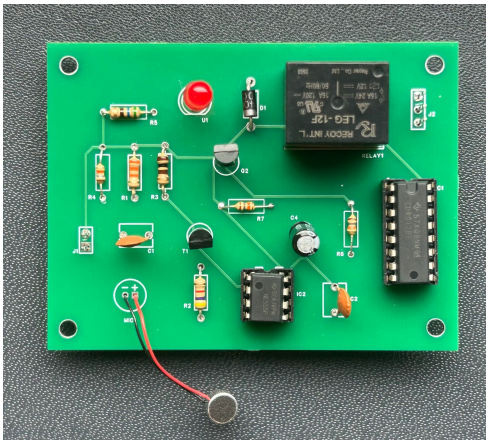
Once the identification of the components was completed, they were inserted into the PCB. Resistors were mounted adhering to the surface of the PCB, while components such as LEDs, transistors and capacitors were positioned leaving a small gap to the PCB.

Special Standard DIP sockets were used for the NE555 and CD4017 integrated circuits. This solution also simplifies any component maintenance or replacement operations.

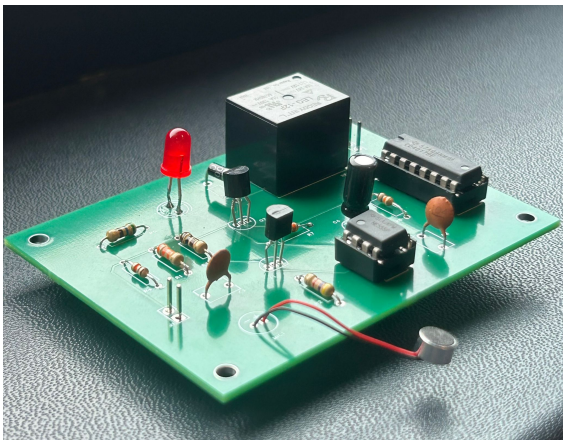
The solder joints were made manually using a soldering iron following the procedure used during laboratory exercises. After the components were inserted, the terminals were slightly bent on the underside of the board to hold them in place during soldering. Subsequently, solder was applied, making both the electrical connection and the mechanical fixing of the components to the PCB.

Particular attention was paid to the orientation of polarized components, such as LEDs, electrolytic capacitors, transistors, diodes and electret microphone, verifying their correct positioning before final soldering.

At the end of the assembly, a visual inspection of the solder joints was carried out to verify the absence of short circuits or incorrect connections. Figure 8 shows the board fully assembled.



(a) Top view of the assembled PCB.



(b) Perspective view of the assembled board.

Figure 8: Clap Switch board fully assembled and ready for testing.

The circuit was subsequently subjected to functional testing in the laboratory. Power was supplied via an AXIOMET AX-3005D laboratory DC power supply, set to a voltage of approximately 10.3 V and connected to the J1 input terminals of the PCB.

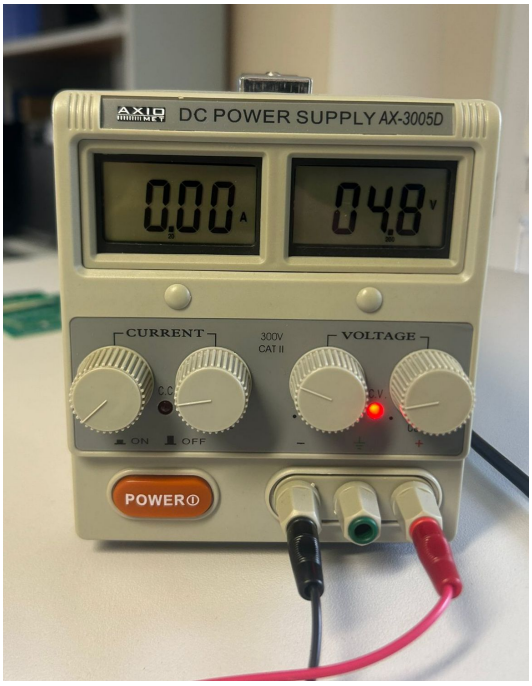
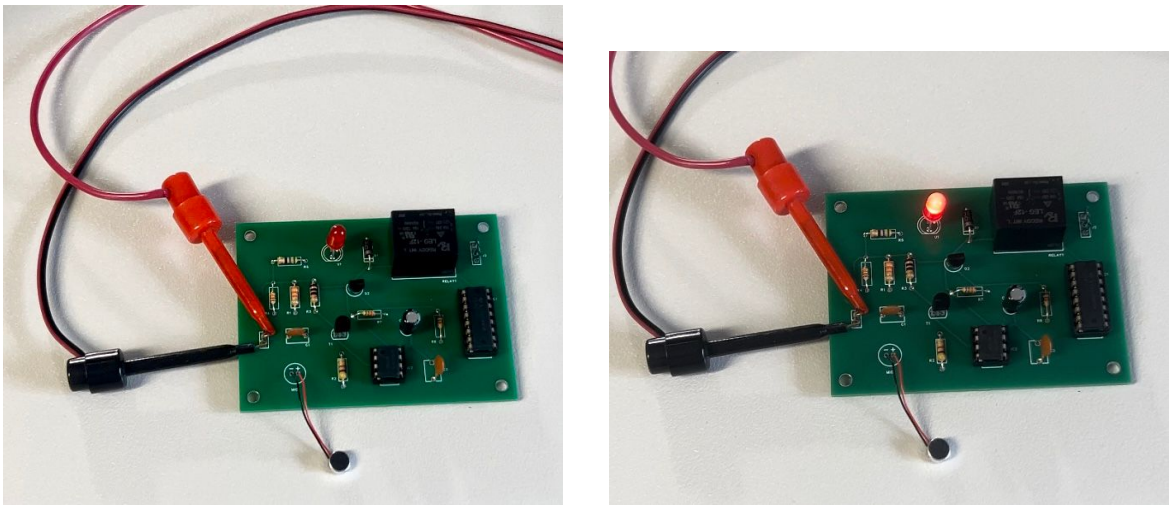


Figure 9: AXIOMET AX-3005D DC Power Supply used during circuit testing.

The operation was verified by applying handclaps near the microphone. From the first tests it was possible to observe the correct functioning of the control logic thanks to the characteristic switching noise produced by the relay, clearly audible with each change in state.

During testing, however, abnormal behavior of the signal LED was found, which remained off even though the relay switched correctly. Although the LED was mounted following the directions of the footprint on the PCB, it was decided to reverse its orientation. After this change the LED started working properly and the system responded as expected.

The final test confirmed the correct functioning of the circuit: with each clap of hands the relay switched its state and the LED turned on or off consistently with the output state. The results obtained therefore validated both the electrical diagram and the PCB design and the assembly operations carried out.



(a) Initial state with LED off. *(b) Status after clapping with LED on.*

Figure 10: Frame of before and after clapping hands during the test.

10 Conclusions

The project enabled the successful development and construction of a Clap Switch-type electronic circuit, capable of controlling the switching on and off of a load with a simple clap of hands.

During the work, all the main phases of the electronic design process were addressed: from the analysis of the initial need to the creation of the electric diagram, from the selection of components and footprints to the design of the PCB, up to the production, assembly and testing of the board.

The experimental tests carried out confirmed the correct functioning of the system and the validity of the design choices adopted. The project also built on practical expertise in using EasyEDA, printed circuit board design, and electronic component assembly and soldering techniques.

The experience provided represented a concrete application of the knowledge acquired during the course, highlighting the importance of a methodical and accurate approach at all stages of electronic device development.

Acknowledgements

I would like to thank the course teacher, Dr. Inż. Łukasz Ruta, for the support provided during the project development and for the availability demonstrated in the laboratory activities.

The recurring question ”Are you sure?” became, during the exercises, a useful reminder about the importance of carefully checking each phase of the project, from the design of the wiring diagram to the final testing of the realized board.